

MA341

APPLIED ODE'S

Review 2

- 1.** Find $|A|$, A^{-1} , and eigenvalues and eigenvectors of the following matrices

a. $\begin{pmatrix} 5 & 1 \\ 2 & 4 \end{pmatrix}$

b. $\begin{pmatrix} 2 & 2 & -1 \\ 1 & 3 & -1 \\ -1 & -2 & 2 \end{pmatrix}$

c. $\begin{pmatrix} -1 & 0 & 1 \\ -3 & 4 & 1 \\ 0 & 0 & 2 \end{pmatrix}$

- 2.** Solve the following homogeneous problem $x'(t) = \begin{pmatrix} 3 & -2 \\ 4 & -1 \end{pmatrix} x(t)$

- 3.** Solve the following systems of ODEs subject to given initial conditions.

a. $x'(t) = \begin{pmatrix} 2 & -1 \\ 3 & -2 \end{pmatrix} x(t) + \begin{pmatrix} e^t \\ t \end{pmatrix}, x(0) = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

b. $x'(t) = \begin{pmatrix} 5 & 3 \\ -6 & -4 \end{pmatrix} x(t) + \begin{pmatrix} 1 \\ e^t \end{pmatrix}, x(0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

c. $x'(t) = \begin{pmatrix} 2 & -1 \\ 5 & -2 \end{pmatrix} x(t) + \begin{pmatrix} \cos t \\ \sin t \end{pmatrix}, x(0) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$